

ENVS-193DS Final

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Setting up

```
# reading in package: tidyverse
library(tidyverse)
# reading in package: ggplot2
library(ggplot2)
# reading in package: rstatix
library(rstatix)
# reading in package: here
library(here)
# reading in package: readr
library(readr)
# reading in package: janitor
library(janitor)
# reading in package: DHARMA
library(DHARMA)
# reading in package: ggeffects
library(ggeffects)
```

```
# reading in nests data using here package
nests <- read_csv(here("data", "nest_data_final.csv"))
```

```
# reading in my data using here package
my_data <- read_csv(here("data", "personal_data.csv"))
```

Problem 1: Research writing

a. Transparent statistical methods

- In part 1, my coworker used a linear model, as the response variable is the probability of American Avocet microhabitat use and the predictor variable is the distance to water edge.
- In part 2, my coworker used a chi-square test of independence because they were testing for an association between two categorical variables: bird species and habitat type.

b. Suggestions for rewriting

- Part 1: American Avocets were more likely to use some microhabitats than others depending on their distance from the water's edge. This shows that proximity to water is an important variable influencing where American Avocets occupy habitat.
- Part 2: The three bird species (American Avocet, Forster's Tern, Black-necked Stilt) did not use habitat types in the same way. Habitat use patterns differed among species, showing that different shorebirds may prefer or rely on different habitats within the San Francisco Bay.

c. Further analysis

One additional variable that could influence American Avocet microhabitat use is water depth (cm), as different birds prefer varying water depths depending on their characteristics. This could be measured at each survey location using a measurement tool, such as a meter stick, or by utilizing a water-depth gauge. In this specific context, water depth may affect habitat use because American Avocets feed by sweeping their bills through shallow water, so they may preferentially use areas with depths that maximize foraging efficiency.

Problem 2: Data Analysis

a. Clean and wrangling data

```
# creating new object for cleaned nests data
nests_clean <- nests |>
  # keep only columns of interest: case control, height, and ant species
  select(case_control, height_cm, ant_species) |>
  # keep only focal ant species
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  # new column to create scientific names for each ant species
  mutate(
    ant_scientific = case_when(
      ant_species == "AB" ~ "Crematogaster sjostedti",
      ant_species == "RRB" ~ "Crematogaster mimosae",
      ant_species == "BBR" ~ "Crematogaster nigriceps"
    ),
    # reordering ant species to be in given order as stated on final
    ↪ assignment instructions
    ant_scientific = factor(
```

```

ant_scientific,
levels = c(
  "Crematogaster sjostedti",
  "Crematogaster mimosae",
  "Crematogaster nigriceps"
)))

```

b. Response variable

The response variable labeled case control indicates whether a tree contained a bird nest. A value of 1 represents a tree occupied by a bird nest, while a value of 0 represents a tree that did not contain a nest.

c. Purpose of study

- Tree height: In this study, tree height is a continuous quantitative variable measured in centimeters. Taller trees may have a greater probability of nest occurrence because they may provide increased protection from predators, greater visibility, or preferred nesting habitat, all depending on the species and their preferences.
- Acacia ant species: In this study, acacia ant species is a categorical variable with three levels: *C. sjostedti*, *C. mimosae*, and *C. nigriceps*. Nest occurrence may differ among ant species because the ants vary in aggressiveness in their interactions with nesting birds. Consequently, a parent bird may alter their nesting site to avoid certain ant species that are more aggressive.

d. Table of models

```

# creating data frame that summarizes models
model_table <- data.frame(
  # column containing model names: we are making two models (model 1 and
  ↪ model 2)
  Model = c("Model 1", "Model 2"),
  # the response variable for both models is nest occurrence (whether or not a
  ↪ nest occurs)
  Response = c(
    "Nest occurrence",
    "Nest occurrence"),
  # the predictors for both models are tree height and ant species present at
  ↪ a given tree of interest

```

```

Predictors = c(
  "Tree height + ant species",
  "Tree height + ant species"),
# interaction terms: one model with no, one model with yes, as specified by
↪ final assignment instructions
Interaction = c(
  "No",
  "Yes"))

# creating table from the data frame that can pop up in my quarto doc and
↪ look amazing and organized
knitr::kable(
  # using the model_table data frame created above
  model_table,
  # the table title so we can stay organized and know what the table is
  ↪ trying to convey to us
  caption = "Candidate linear models used to evaluate the effects of tree
  ↪ height and acacia ant species on nest occurrence.")

```

Table 1: Candidate linear models used to evaluate the effects of tree height and acacia ant species on nest occurrence.

Model	Response	Predictors	Interaction
Model 1	Nest occurrence	Tree height + ant species	No
Model 2	Nest occurrence	Tree height + ant species	Yes

e. Running the models

```

# model 1: no interaction; glm because binary response variable
mod1 <- glm(
  # assessing case control (nest or no nest) and tree height and ant species
  case_control ~ height_cm + ant_scientific,
  # from clean nests data frame
  data = nests_clean,
  # response variable is binary
  family = binomial)

# model 2: interaction; glm because binary response variable
mod2 <- glm(

```

```
# assessing case control (nest or no nest) and tree height and ant species
case_control ~ height_cm * ant_scientific,
# from clean nests data frame
data = nests_clean,
# response variable is binary
family = binomial)
```

f. Selecting the best model

```
AIC(mod1, mod2)
```

	df	AIC
mod1	4	258.7430
mod2	6	237.8042

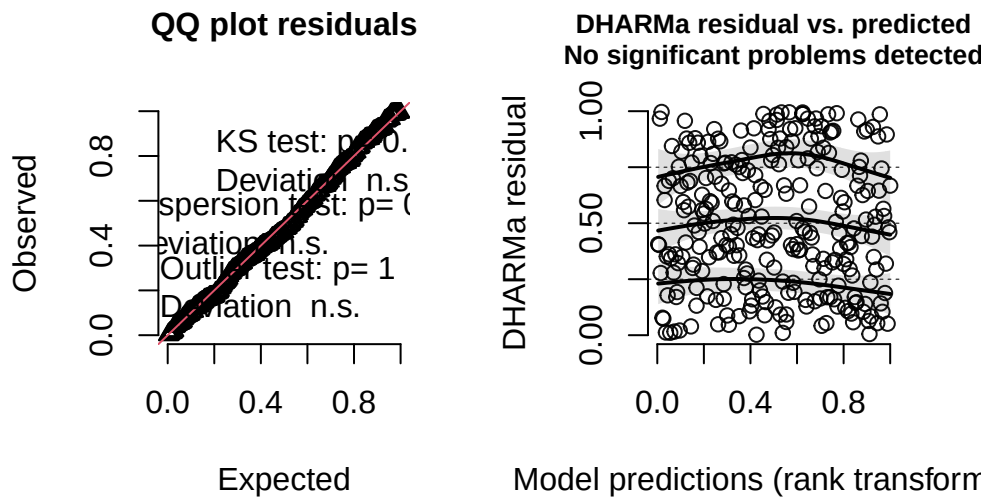
The best model as determined by Akaike's Information Criterion (AIC) included tree height, acacia ant species, and the interaction between tree height and acacia ant species. This model had the lowest AIC value (248.15) compared to the model containing only the main effects of tree height and ant species and no interaction term (263.75).

g. Check the diagnostics

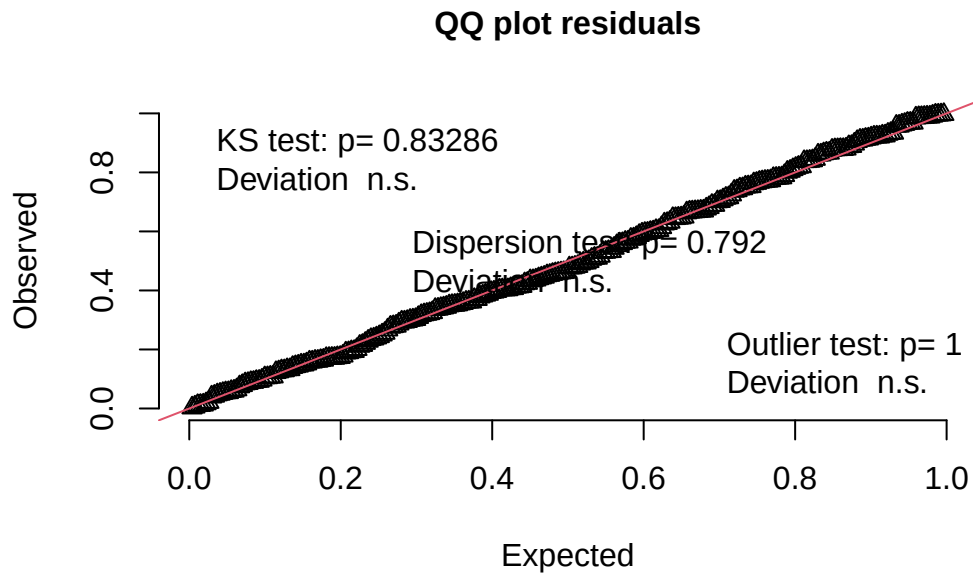
```
sim_res <- simulateResiduals(mod2)

plot(sim_res)
```

DHARMa residual



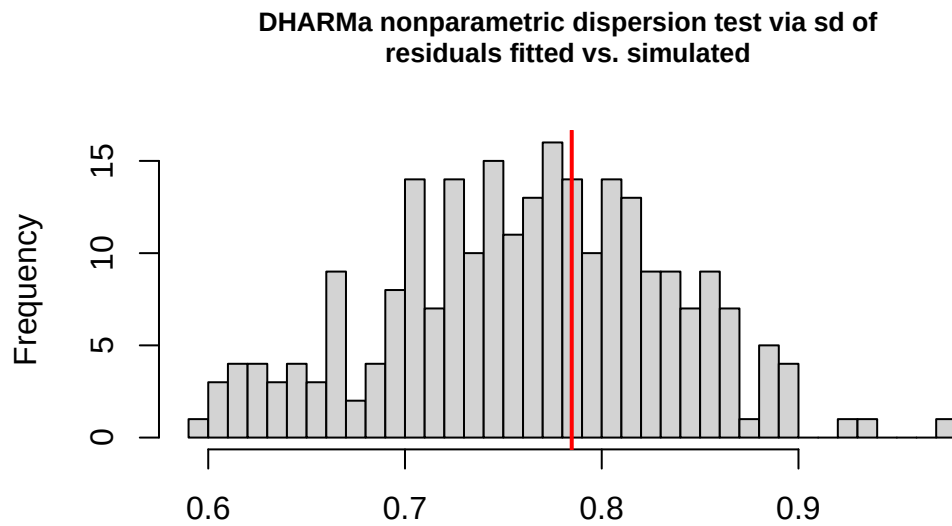
```
testUniformity(sim_res)
```



Asymptotic one-sample Kolmogorov-Smirnov test

```
data: simulationOutput$scaledResiduals  
D = 0.037897, p-value = 0.8329  
alternative hypothesis: two-sided
```

```
testDispersion(sim_res)
```

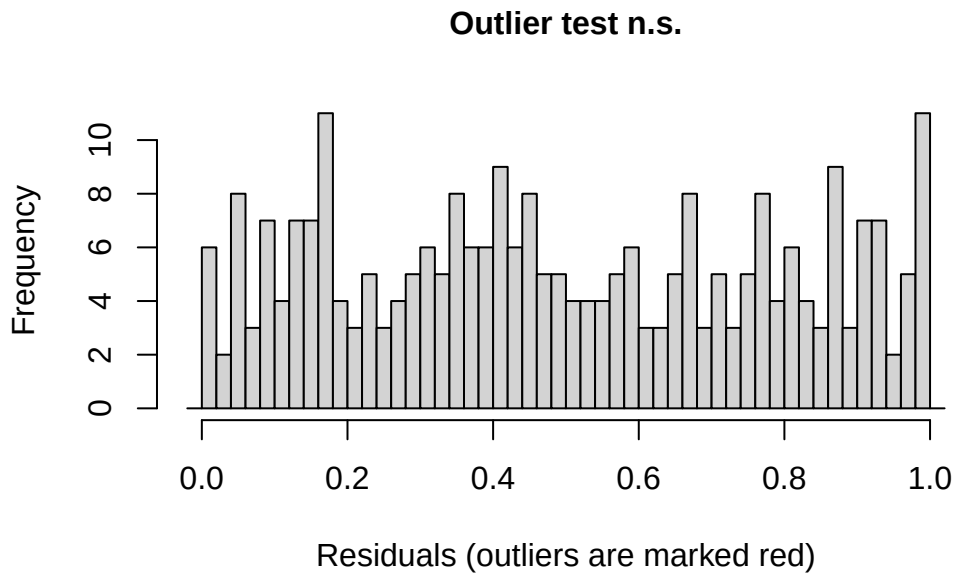


Simulated values, red line = fitted model. p-value (two.sided) = 0.792

DHARMA nonparametric dispersion test via sd of residuals fitted vs.
simulated

```
data: simulationOutput  
dispersion = 1.0284, p-value = 0.792  
alternative hypothesis: two.sided
```

```
testOutliers(sim_res)
```



DHARMA bootstrapped outlier test

```
data:  sim_res
outliers at both margin(s) = 0, observations = 270, p-value = 1
alternative hypothesis: two.sided
percent confidence interval:
 0.000000000 0.001944444
sample estimates:
outlier frequency (expected: 0.000111111111111111 )
0
```

h. Visualizing model predictions

```
# creating predictions object and calculating these with ggpredict
preds <- ggpredict(
  # model 2 was best fit
  mod2,
  # looking at tree height and ant species
  terms = c("height_cm", "ant_scientific"))
```

```

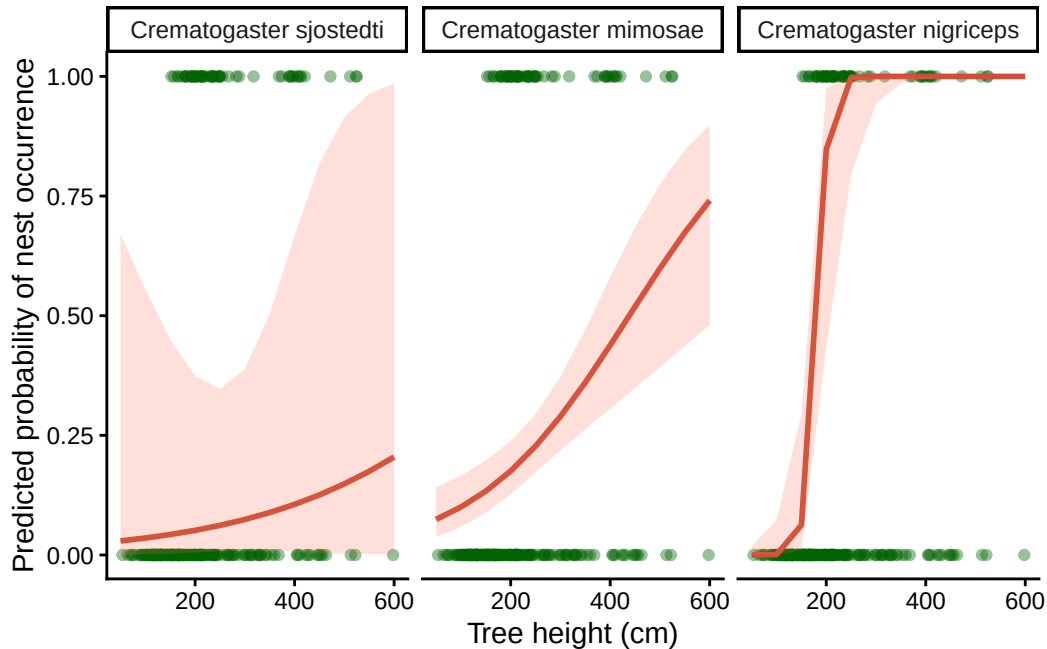
# base layer ggplot with predictions data
ggplot(data = preds,
       # aesthetics x = x and y = predicted
       mapping = aes(x = x,
                     y = predicted)) +
# first layer geom_point for individual observations
geom_point(
  # data cleaned nests data frame
  data = nests_clean,
  # setting axes: x is height, y is case control
  aes(
    x = height_cm,
    y = case_control
  ),
  # making the points dark green
  color = "darkgreen",
  # making the points slightly transparent so you can see all of them
  alpha = 0.4
) +
# second layer geom_line for prediction line
geom_line(
  # preds data frame
  data = preds,
  # setting axes: x is x, y is predicted
  aes(
    x = x,
    y = predicted
  ),
  # color tomato for easy visibility
  color = "tomato3",
  # line width of one for aesthetics
  linewidth = 1
) +
# third layer: geom ribbon makes the confidence interval ribbon around each
↪ one
geom_ribbon(
  # data frame is predictions data frame
  data = preds,
  # setting axes: x is x
  aes(
    x = x,
    # y min is the lower value of the confidence interval
    ymin = conf.low,

```

```

    # y max is the higher value of the confidence interval
    ymax = conf.high
  ),
  # fill in with lighter tomato color for contrast from line
  fill = "tomato1",
  # making it see through so you can see the lines and points through the
  ↪ ribbon line
  alpha = 0.2
) +
# grouping by ant species
facet_wrap(~group) +
# setting titles
labs(
  # x axis title is tree height measured in cm
  x = "Tree height (cm)",
  # y axis title is predicted probability of nest occurrence
  y = "Predicted probability of nest occurrence"
) +
# classic theme for easy readability
theme_classic() +
# removing legend because i find it unnecessary here
theme(
  legend.position = "none"
)

```



i. Writing a caption

Figure 1. Predicted probability of bird nest occurrence as a function of tree height and acacia ant species. Lines represent predictions from the selected linear regression model and shaded ribbons represent 95% confidence intervals. Points indicate observed nest presence (1) and absence (0) for individual trees. Separate panels show predictions for *Crematogaster sjostedti*, *Crematogaster mimosae*, and *Crematogaster nigriceps* ant species. Data accessed from Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.8373322>

j. Calculating model predictions

```
# creating new data frame more predictions
morepreds <- data.frame(
  # we want to find the predictions at 600 cm height tree
  height_cm = c(600, 600, 600),
  # for each of these three ant species:
  ant_scientific = factor(
```

```

      c(
        "Crematogaster sjostedti",
        "Crematogaster mimosae",
        "Crematogaster nigriceps"
      ),
      levels = levels(nests_clean$ant_scientific)
    )
  )

# using predict function to find our predictions for this step
predict(
  # using model 2
  mod2,
  # data is more preds data frame we just made
  newdata = morepreds,
  # response
  type = "response"
)

```

```

           1           2           3
0.2048017 0.7407857 1.0000000

```

k. Interpreting results

The probability of nest occurrence varied with both tree height and acacia ant species (Figure 1). For a tree height of 600 cm, the predicted probability of nest occurrence was 0.205 for trees occupied by *Crematogaster sjostedti*, 0.741 for trees occupied by *Crematogaster mimosae*, and about 1.000 for trees occupied by *Crematogaster nigriceps*. In other words, a 600 cm tree occupied by the *Crematogaster nigriceps* ant species is going to have a nest. The relationship between tree height and ant species shows that the effect of tree height on nest occurrence were different among ant species, with taller trees occupied by *C. nigriceps* showing the greatest probability of nest occurrence. According to Lujan, Nielsen, et al. (2023), differences in nest occurrence among ant species may be explained by variation in ant defensive behavior, as some acacia ants provide protection from predators while being less disruptive to nesting birds. These results suggest that both tree characteristics and ant species that already live in that tree do, in fact, influence bird nest site selection.

Problem 3: Affective and exploratory data

a. Comparing visualizations

- My visualizations were very different from each other in the way that I represented my data. My exploratory visualizations focused more on showing visual patterns and trends between variables in my data. My affective visualization did that as well; however, it did it in a much more creative way, as it told a story about my entire day using my data as a tool to do so. There was much more story-telling involved in it.
- All of my visualizations, both exploratory and my affective visualization, effectively demonstrated patterns regarding my water intake for the day and the conditions I experienced. My exploratory visualizations explicitly showed these through easy-to-read visualizations, while my affective visualization provided the viewer with multiple daily statistical summaries of my personal data that they could then compare.
- Generally, I noticed in my exploratory visualizations that every variable I considered at least had some solid influence on my daily water intake. For my affective visualization, however, the main pattern shown is the fact that my water intake varies from day to day. So although these sound similar, one kind of visualization (exploratory) explicitly laid out the fact that the variables had relationships, the other kind of visualization (affective) showed the viewer the larger picture of the idea that my water intake is not a fixed value that remains the same every day. I believe this is because I created my affective visualization primarily with the intention of telling a story about my day and how heat and activity contributes to my water intake, rather than focusing on explicitly demonstrating how these variables vary day-by-day along with my water intake.
- I received excellent feedback from my peers! One of them suggested that I add in a couple more days of data into my visualization, which I did end up doing in the form of old posters laying on the floor. However, I still chose to highlight one main day of data, as I found it easier to tell a story about a single day by utilizing primarily that day's data. I basically added the other days so viewers could compare my water intakes depending on the day (I was inspired by calendar pages), but in the end, I chose to focus on a "day-in-my-life" narrative.

b. Designing an analysis

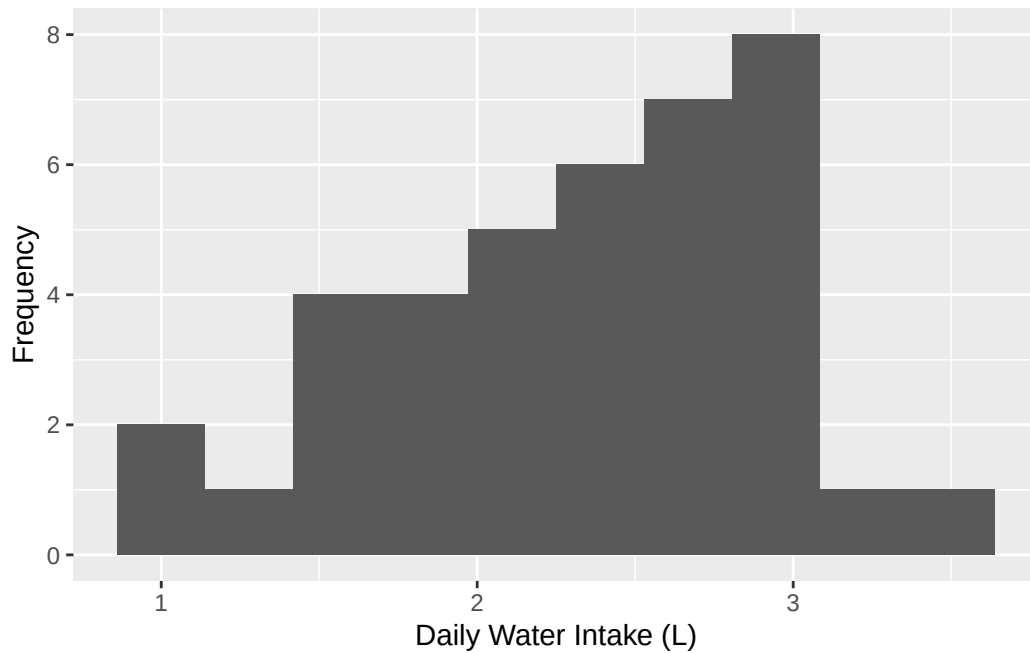
To determine whether daily water intake differed between days when I was sick and days when I was not sick, I am using a Wilcoxon rank-sum test. This analysis is appropriate because daily water intake (measured in L) is my response variable and sickness (sick or not sick) is a categorical predictor variable with two levels.

Side note: My personal prediction is that there is a correlation, and that I tended to drink more water on days I was sick than days I was not.

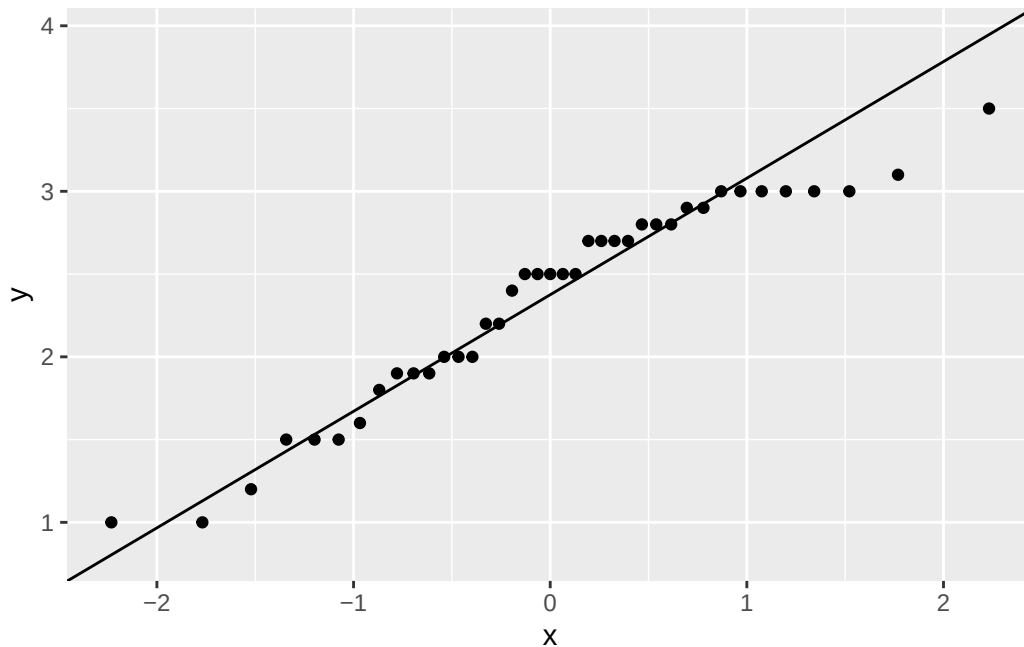
c. Checking assumptions and running an analysis

```
# creating new object for cleaned data using my_data data frame
my_data_clean <- my_data |>
  # cleaning names: all lowercase and adding underscores
  clean_names() |>
  # removing an empty row i accidentally included
  remove_empty("rows") |>
  # converts sick status into categorical variable
  mutate(
    sick = factor(sick_y_n)
  )
```

```
# looking at normal distribution using histogram
# base layer ggplot with cleaned data frame
ggplot(data = my_data_clean,
  # setting x axis: water intake in liters
  mapping = aes(x = water_intake_l)) +
  # first layer geom_histogram with 10 bins
  geom_histogram(bins = 10) +
  # relabeling x and y axis
  labs(x = "Daily Water Intake (L)", # x axis is daily water intake in liters
    y = "Frequency") # y is frequency (number observations in each bin)
```



```
# looking at distribution using a qq plot
# base layer ggplot with cleaned data frame
ggplot(data = my_data_clean,
        # looking at water intake in liters
        mapping = aes(sample = water_intake_l)) +
# adding individual observations
stat_qq() +
# adding in reference line to assess normality of observations
stat_qq_line()
```



I examined the distribution of daily water intake using a histogram and a Q-Q plot. The histogram indicated that water intake was left-skewed, and the Q-Q plot showed deviations from the reference line in the tails, suggesting that the data were not normally distributed. Because the assumption of normality was not met, a Wilcoxon rank-sum test is still appropriate to compare water intake between sick and non-sick days.

```
# creating new object from running the wilcoxon rank-sum test
wilcox_results <- wilcox.test(
  # examining correlation between water intake and sickness
  water_intake_l ~ sick,
  # using my cleaned data data frame
  data = my_data_clean
)

# displaying results
wilcox_results
```

Wilcoxon rank sum test with continuity correction

```
data: water_intake_l by sick
W = 50.5, p-value = 0.02497
alternative hypothesis: true location shift is not equal to 0
```

```
# calculating effect size
effect_size <- wilcox_effsize(
  # my cleaned data frame
  my_data_clean,
  # examining correlation between water intake and sickness
  water_intake_l ~ sick
)

# displaying results
effect_size
```

```
# A tibble: 1 x 7
  .y.      group1 group2 effsize    n1    n2 magnitude
* <chr>    <chr>  <chr>    <dbl> <int> <int> <ord>
1 water_intake_l N      Y      0.362   32    7 moderate
```

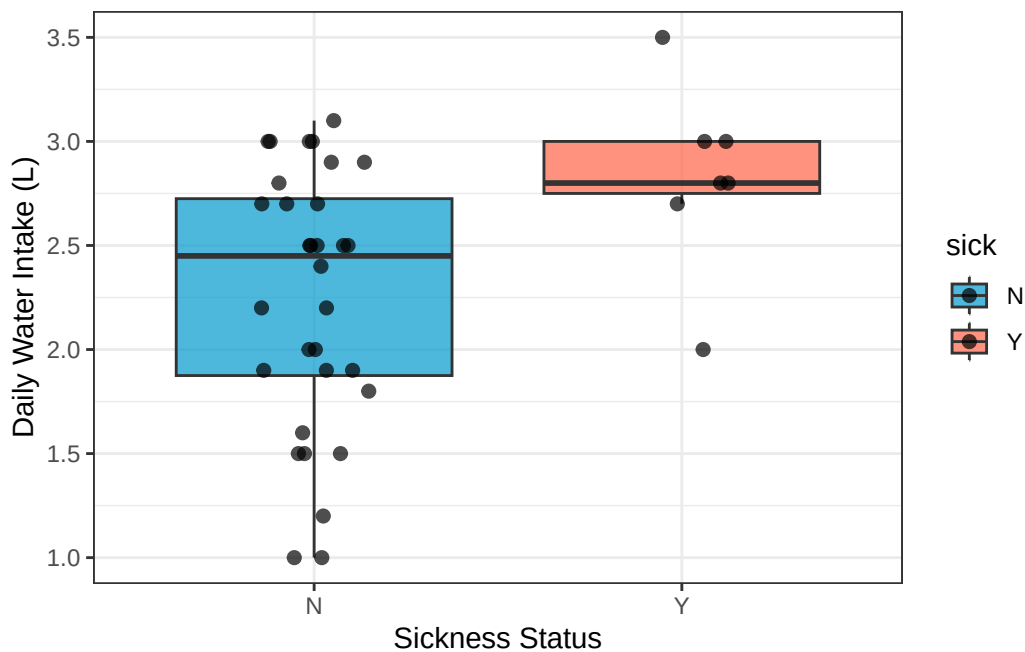
d. Create a visualization

```
# base layer ggplot with cleaned data frame
ggplot(data = my_data_clean,
  # setting x and y axes
  mapping = aes(x = sick, # x axis is sickness
    y = water_intake_l, # y axis is water intake in liters
    # fill color by sickness yes or no
    fill = sick)) +
# first layer boxplot
geom_boxplot(alpha = 0.7, # alpha 0.7 so you can still see individual
  ↪ observations through boxplot
  # taking out extreme outliers that may mess things up
  outlier.shape = NA) +
# second layer jitterplot to show individual observations
geom_jitter(width = 0.15, # controlling width
  size = 2, # size 2 for easy readability
  height = 0, # controlling height
  alpha = 0.7) + # alpha 0.7 so you can see boxplot too
# setting custom colors
scale_fill_manual(values = c("deepskyblue3", # not sick = pretty blue
  "tomato1")) + # sick = red
# setting axes titles
labs(
  # x axis is sickness status (sick or not sick)
```

```

x = "Sickness Status",
# y is daily water intake measured in liters
y = "Daily Water Intake (L)"
) +
# choosing a nice, simple theme
theme_bw()

```



e. Figure caption

Figure 1. Daily water intake on days when I was sick and days when the I was not sick. Boxplots show the median and interquartile range of daily water intake (measured in liters) for each sickness category (yes and no). Individual observations are displayed as black jittered points to show the distribution of measurements. The data around the blue box indicates days when the participant was not sick and those around light red box indicates days when the participant was sick.

f. Write about your results

Daily water intake differed significantly between days when I was sick and days when I was not sick (Wilcoxon rank-sum test, $W = 50.5$, $p = 0.025$). As shown in Figure 1, the median water intake was higher on sick days than on non-sick days, supporting my prediction that I would

drink more water when ill. Sickness status was associated with a moderate effect on water consumption ($r = 0.362$), suggesting that the difference was not only statistically significant but also meaningful in magnitude.